RX INTERVIEWS

BOND MATH QUESTIONS

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As you know if you've gone through the other guides, a relatively common type of restructuring interview question revolves around finding the YTM when given the coupon rate, years to maturity, and current trading price.

But sometimes you'll get more conceptual "bond math" questions where your interviewer wants you to eschew doing any "real" math (e.g., using the estimated YTM formula) and explain things more extemporaneously.

To that end, I've put together some questions below that will hopefully help prompt your thinking in the right direction and have tried to explain things in a more generalized way even if that's at the cost of glazing over some nuance.

However, if you're short on time and long on interview-related stress, the only question that routinely comes up is the second question below, and you can answer it with either the estimated YTM formula or the more "intuitive" approach discussed in the answer. The final question here will also occasionally come up although the right (interview appropriate) approach to answering it should be clear after reading the second question.

Just keep in mind that whenever you have a bond math question in an interview you'll never actually be expected to do any convoluted calculations or need to explain any more cumbersome concepts (e.g., convexity). Instead, everything will usually be kept quite surface level and your interviewer is normally just looking for you to demonstrate a handwavy, directional understanding of what's going on.

Anyway, to set the stage a bit, there are a number of different ways you can think about framing what a YTM is *really* saying. The most common (academic) framing is that it's the rate at which you can discount the future cash flows (coupons and principal) of a bond so that the present value of those cash flows ends up aligning with the current price.

So, for example, if you have a bond with four years to maturity, a 5% coupon, and a yield of 7% then you know the current price should be 92.23. Below you can see that if you just discount the cash flows you're getting (e.g., $5 in coupons a year, $100 at maturity) using the yield as your discount rate then you'll end up with a PV of 93.23. Alternatively, if we pretend the coupon is equal to the yield (e.g., both 5%) we get a PV of 100 as we'd expect.

Note: If we're holding the bond to maturity and reinvesting the coupons clipped at the YTM rate then our return (IRR) should equal the YTM, which is also what we find below.

Note: Throughout we'll be assuming that we're getting coupons annually, and you can use this as your YTM calculator: https://dqydj.com/bond-yield-to-maturity-calculator/

Note: It's always important, especially when dealing with more nuanced topics, to have an idea of what your interviewer *wants* and *expects* to hear. So, don't get too lost in the sauce here and focus most on the second question and the last question where I make it clear how you should approach answering in an interview. The other questions primarily involve me attempting (hopefully successfully) to make this all slightly more intuitive.

Assumptions	
Maturity	4.0
Coupon	5.0%
Yield	7.0%
Price	93.23

PV Cash Flow Method	Year 1	Year 2	Year 3	Year 4
Coupons	5.00	5.00	5.00	5.00
Principal	-	-	-	100.00
Total Cash Flows	**5.00**	**5.00**	**5.00**	**105.00**
PV of Total Cash Flows	**4.67**	**4.37**	**4.08**	**80.10**
PV if C = YTM	**4.76**	**4.54**	**4.32**	**86.38**

PV of Total Cash Flows	**93.23**
PV of Cash Flows if C = YTM	**100.00**

IRR Calculation	Year 0	Year 1	Year 2	Year 3	Year 4
	(93.23)	5.00	5.00	5.00	105.00
IRR	**7.0%**				

This is all well and good. But it perhaps doesn't build that much of an intuitive understanding – beyond illustrating how as the YTM increases the price decreases – so let's see if we can circle around a few more creative ways of thinking about all of this.

- **Let's imagine we have a bond with four years to maturity, a 5% coupon, and a 7% yield that's trading at 93.2 and will return par. How should the price of the bond change as it approaches maturity and how does the yield intuitively inform it?**

This isn't the kind of question that'd actually come up in an interview but since you'll occasionally get more conceptual bond math questions – that get beyond just using the estimated YTM formula – let's use this question as an excuse to take a step back for a second, eschew doing any real math, and think about bond pricing more intuitively.

Whenever you depart from relying on real math and begin trying to explain things more generally, you'll inherently lose some nuance. But here's one way you can think about yield in relation to bond pricing at a (maybe!) more intuitive level: it's reflecting the amount the bond needs to return in a given year, based on its *current price* in a given year, to get it on a pathway towards being worth par at maturity.

This is something that many initially overlook, even though it's obvious in retrospect: if you have a bond trading below par with a number of years until maturity and we say the yield is X%, then the value of that bond must increase as maturity approaches if the yield is to stay static through the period. Because if the bond weren't to increase in value then the natural consequence would be for the yield to inch upwards as maturity approaches.

To put some meat on the bones, in our little example here we've been told a bond has four years to maturity, a coupon of 5%, and a yield of 7%. Consequently, the price of the bond will be around 93.2 assuming a par value of 100 (assume annual coupons).

If we were to plot the four years until maturity, what we should observe is that the price of the bond increases incrementally each year as we approach maturity until it reaches par while the actual yield stays static.

The obvious question then is how much should we theoretically expect the bond to increase each year. To answer that, let's think about what yield is representing once again: it's the return of the bond necessary to get it back to being worth par at maturity. Therefore, for a bond trading below par, that'll necessitate there being a return in excess of the coupon rate because the price of the bond needs to "catch up" to the par value.

So, the theoretical amount the bond should *increase* in value in a given year should be reflective of the dollar (not percentage) spread between the return of the bond in a given year, based on the current price, and the coupon (which will obviously be a static value).

If this all seems a bit fuzzy, think about the little table below…

Assumptions	
Maturity	4.0
Coupon	5.0%
Yield	7.0%
Price	93.23

	Year 1	**Year 2**	**Year 3**	**Year 4**
Starting Bond Price	93.23	94.75	96.38	98.13
Plus: Yield	6.53	6.63	6.75	6.87
Less: Coupon	(5.00)	(5.00)	(5.00)	(5.00)
Final Bond Price	**94.75**	**96.38**	**98.13**	**100.00**
Appreciation Spread (AS)	*1.53*	*1.63*	*1.75*	*1.87*
YTM Calculation Check	*7.0%*	*7.0%*	*7.0%*	*7.0%*
Actual Ending Price Check	94.75	96.38	98.13	100.00

We begin by taking our purchase price of 93.23 (based on there being a coupon of 5%, a yield of 7.0%, and four years to maturity). Applying the yield of 7.0% to the purchase price, that means the bond itself, in year one, "earns" a return of around 6.53. Stripping out the coupon component paid, this means the bond has 1.53 of price appreciation, and should theoretically be worth 94.751.

Now if you were to go use a yield to maturity calculator and type in a coupon of 5%, three (not four!) years to maturity, and a current trading price of 94.751 you'd find the yield to maturity is, as you'd guess, equal to 7.0%. Likewise, if you just play around with a YTM calculator and try to get a YTM of 7.0% by rejiggering prices around, you'll hone in on, for example, 96.384 if you put a coupon of 5% and say there are just two years to maturity.

If you then tally up what I'm calling here the "appreciation spread" for all four years (e.g., the spread between the amount of return in a given year after stripping out the coupon component) you'll get 6.77. And, since our initial purchase price was 93.23, adding these values together gets you 100 at maturity which must be the value of the bond at maturity.

Therefore, we've proven that the final value of the bond should be par (100) if it grows by 7% a year and, from an investor's perspective, each year your effective return from owning the bond is per se going to be informed by having bought the bond below par.

So, glossing over some nuance, the takeaway here is that one way to think about yield when a bond is trading below par, from a slightly different perspective, is that it's reflecting the return necessary for a bond to reach a par value at maturity and that it will, obviously, be an amount in excess of the coupon rate as it's trying to "catch up" to par.

- **Let's say we have a bond maturing in five years with a 10% coupon that's currently trading at 80 and will return par. Can you tell me, using as little math as possible, what the YTM roughly is?**

Here we have all the ingredients needed to use the estimated YTM formula and, if we were to use it, we'd end up quickly getting 14 / 90 or 15.56%.

But let's pretend that your interviewer balks at your utilization of the estimated YTM formula and says they want you to give a more "intuitive" answer. In this case, hopefully the rather lengthy question above helps prompt your thinking in the right direction.

Because what we found through the prior question is that a simplified way we can think about a bond trading below par is that its YTM is being informed by two components: the coupon component and the price appreciation component.

So, the way your interviewer will want you to answer this question is by thinking about these components separately…

First, we know that our YTM will be driven partly by the coupons we're clipping (in this case, $10 a year). But how favorable those coupons look is inherently contingent on the price that the bond is *currently* trading at (e.g., what you're paying to get these coupons).

Therefore, a way to think about the portion of our YTM being driven by the coupons we're clipping, and their subsequent reinvestment until maturity at the YTM rate, can be represented by the current yield (C / P) or, in this case, 10/80 = 12.5%.

Second, we know that our YTM will be driven partly by the price appreciation that's occurring since we're buying the bond below par and assuming it returns par in five years. As we saw in the prior example, the amount the bond will increase per year won't be constant, but the amount of price appreciation each year, when tallied together, will equal the spread between the purchase price and par. In other words, the bond will reach par at maturity, it just won't be going up each year by the same amount.

Therefore, a simplified way we can estimate the portion of our YTM that's being driven by price appreciation is that it's just the spread between par and the purchase price divided by the number of years until maturity.

In this case, it'll be (100-80) / 5 = 4. So, we can say that we're getting roughly 4% worth of principal pickup (price appreciation) per year.

Adding these two YTM components together we get 12.5% + 4% or 16.5%. Needless to say, this approach isn't really much quicker than using the estimated YTM formula – but it does allow you to show a more intuitive understanding of what's driving the YTM.

Assumptions	
Maturity	5.0
Coupon	10.0%
Par Value	100.0
Price	80.0

Estimated YTM	
Current Yield (C / P)	12.5%
Appreciation Adjustment (%)	4.0%
Estimated YTM	**16.5%**

The real YTM here is 16.126% (assuming annual coupons) and, if you're curious, you can get a feel for how this bond changes as maturity approaches by revamping our quick-and-dirty analysis from the prior question…

Assumptions	
Maturity	5.0
Coupon	10.0%
Yield	16.1%
Price	80.0

	Year 1	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Price	80.00	82.90	86.27	90.18	94.72
Plus: Yield	12.90	13.37	13.91	14.54	15.28
Less: Coupon	(10.00)	(10.00)	(10.00)	(10.00)	(10.00)
Final Price	**82.90**	**86.27**	**90.18**	**94.72**	**100.00**
Appreciation Spread (AS)	*2.90*	*3.37*	*3.91*	*4.54*	*5.28*
Current Yield (C / P)	12.5%	12.1%	11.6%	11.1%	10.6%
App. Spread Yield (AS / P)	3.6%	4.1%	4.5%	5.0%	5.6%
Yield to Maturity	**16.1%**	**16.1%**	**16.1%**	**16.1%**	**16.1%**

- **Let's say that we have two bonds. The first matures in five years with a 10% coupon and is trading at 80. The second matures in four years with a 10% coupon and is also trading at 80. What one do you think has a higher YTM?**

If we use our strategy from the question above – eschewing the "real math" of the estimated YTM formula in favor of a more "intuitive" approach – then we'll find that the first bond has a YTM of 16.5% while the second bond has a YTM of 17.5%.

In both cases the current yield (C/P) is identical. But in the case of the former we're getting $20 in price appreciation in five years, whereas in the latter we're getting it in just four years.

So, it makes perfect sense that the shorter-duration bond should have a higher YTM because, all else being equal, it's obviously better to get the same amount of cash sooner rather than later.

Assumptions	
Maturity	4.0
Coupon	10.0%
Par Value	100.0
Price	80.0
Estimated YTM	
Current Yield (C / P)	12.5%
Appreciation Adjustment (%)	5.0%
Estimated YTM	**17.5%**

Assumptions	
Maturity	5.0
Coupon	10.0%
Par Value	100.0
Price	80.0
Estimated YTM	
Current Yield (C / P)	12.5%
Appreciation Adjustment (%)	4.0%
Estimated YTM	**16.5%**

- **Let's say that I didn't believe that YTM actually assumes coupons are being reinvested at the YTM rate. How could you try to convince me it's actually true?**

This wouldn't be asked in an interview context as it's a little too conceptual. But working through this question will (hopefully!) allow us to put a bow on all of this yield talk and give you a reasonably intuitive understanding of the basics.

Note: In the future maybe I'll put together a few questions on thinking through duration and convexity. But we'll save that for a later day since no one is going to ask you the intuition behind convexity declining as yields increase in an interview.

Anyway, one way to show a coupon-reinvesting-skeptic the error of their ways would be to lay out the cash flows: take your coupons each year, compound them at the YTM rate for the relevant number of years left until maturity, tack on the principal you're getting at maturity, and sum up the proceeds.

Then you could compare the total proceeds found to if you had just placed an amount of money, equivalent to the current trading price of the bond, into a bank account that promised to pay you an annual rate equivalent to the YTM for the number of years until maturity.

In both instances you'll find that the total proceeds are equivalent and that their present value (when using the YTM as the discount rate) will bring you back to the current price.

Further, your return (IRR) will be the same whether you're assuming you get the coupons each year and par at maturity or if you're assuming you get no coupons and a lump sum of all the proceeds at maturity (keep in mind with the lump sum strategy you're assuming that you're taking the coupons each year and immediately locking them into an investment paying the YTM rate for however many years until maturity – consequently, all proceeds arrive effectively only at maturity).

In the end, the YTM must assume that the coupons are being reinvested at the YTM rate. Because if the coupons weren't reinvested at all, or were done so at some lower rate, then the PV of our total proceeds in Strategy B would be lower than the current price and the proceeds from Strategy A would be higher than the proceeds from Strategy B.

Note: Both of the IRR figures found below are equivalent to the YTM that we used (as they should be since the YTM should equal the IRR if you're holding the bond to maturity and reinvesting coupons at the YTM rate).

Assumptions	
Maturity	4.0
Coupon	10.0%
YTM	16.5%
Price	82.0

Strategy A (Investment)	
Investment	82.0
Rate	16.50%
Total Proceeds at Maturity	**151.03**

Strategy B (Coupon Reinvestment)	Year 1	Year 2	Year 3	Year 4
Coupons	10.00	10.00	10.00	10.00
Coupon Reinvestment Earnings	5.81	3.57	1.65	-
Principal	-	-	-	100.00
Total Proceeds	**15.81**	**13.57**	**11.65**	**110.00**

Total Proceeds at Maturity	**151.03**
PV of Proceeds at Maturity	**82.00**

IRR Calculation (Annual Coupons)

Year 0	Year 1	Year 2	Year 3	Year 4
(82.00)	10.00	10.00	10.00	110.00

IRR	**16.5%**

IRR Calculation (Lump Sum)

Year 0	Year 1	Year 2	Year 3	Year 4
(82.00)	-	-	-	151.03

IRR	**16.5%**

Just as a point of reference, if we assume that we have a bond trading at par – meaning that the coupon rate and YTM are the same – then, as you'd expect, we get an IRR that's 10% regardless of if we do the lump sum method or assume coupons are paid annually.

Assumptions	
Maturity	4.0
Coupon	10.0%
YTM	10.0%
Price	100.0

Strategy A (Investment)	
Investment	100.0
Rate	10.00%
Total Proceeds at Maturity	**146.41**

Strategy B (Coupon Reinvestment)	**Year 1**	**Year 2**	**Year 3**	**Year 4**
Coupons	10.00	10.00	10.00	10.00
Coupon Reinvestment Earnings	3.31	2.10	1.00	-
Principal	-	-	-	100.00
Total Proceeds	**13.31**	**12.10**	**11.00**	**110.00**

Total Proceeds at Maturity	**146.41**
PV of Proceeds at Maturity	**100.00**

IRR Calculation (Annual Coupons)

Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**
(100.00)	10.00	10.00	10.00	110.00
IRR				
10.0%				

IRR Calculation (Lump Sum)

Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**
(100.00)	-	-	-	146.41
IRR				
10.0%				

- **Let's assume we have a bond maturing in four years with a 10% coupon that's trading at 82. Theoretically, the YTM should be 16.5%. But let's assume we can only reinvest coupons at a lower rate of 5%. Do you think the overall return would be higher or lower?**

Theoretically, if we're reinvesting coupons at the YTM rate, the IRR should be 16.5%. But it's very unlikely, in the real world, that when you get your coupons you're going to be able to immediately turn around and reinvest them at this high of a rate consistently.

In an interview context it'd be impossible to go through the math to find your actual IRR based on a 5% reinvestment rate. But you should have the directional understanding that if you're reinvesting your coupons at this lower rate, then that's going to supress down your overall IRR.

For example, think about the analysis done in the prior question: your coupon reinvestment earnings are going to be less than before, so your total lump sum proceeds will be less. Therefore, your overall IRR will unequivocally be lower – you just can't say by how much in an interview given the math involved in figuring it out.

Strategy B (Coupon Reinvestment)				
	Year 1	**Year 2**	**Year 3**	**Year 4**
Coupons	10.00	10.00	10.00	10.00
Coupon Reinvestment Earnings	1.58	1.03	0.50	-
Principal	-	-	-	100.00
Total Proceeds	**11.58**	**11.03**	**10.50**	**110.00**

Total Proceeds at Maturity	**143.10**

IRR Calculation (Annual Coupons)				
Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**
(82.00)	10.00	10.00	10.00	110.00
IRR	**16.5%**			

IRR Calculation (Lump Sum)				
Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**
(82.00)	-	-	-	143.10
IRR	**14.9%**			

Note: You could also find this same answer by using the total return formula. But that's much too involved (computationally) to ever do in an interview, and there's zero need to memorize the formula.

- **In prior questions we've said that the current yield captures the portion of the YTM stemming from the coupons. But is that really true? Does the current yield really capture not only the coupons but their reinvestment earnings? How would you go about showing that?**

This question most certainly won't come up in an interview, but it'll allow us to recap and intertwine a few of our prior approaches nicely. So, let's say that we're dealing with a bond that has five years to maturity, a coupon of 10%, and a YTM of 16.126%.

As we discussed in the preamble, one way to find the price is just to discount the actual cash flows you're getting (e.g., $10 in coupons annually, $100 at maturity) by the YTM. If we do that, we'll end up getting a price of $80.

PV Cash Flow Method

	Year 1	Year 2	Year 3	Year 4	Year 5
Coupons	10.00	10.00	10.00	10.00	10.00
Principal	-	-	-	-	100.00
Total Cash Flows	**10.00**	**10.00**	**10.00**	**10.00**	**110.00**
PV of Total Cash Flows	**8.61**	**7.42**	**6.39**	**5.50**	**52.09**
PV if C = YTM	**9.09**	**8.26**	**7.51**	**6.83**	**68.30**

PV of Annual Cash Flows	**80.00**
PV of Cash Flows if C = YTM	**100.00**

In a prior question we also convinced ourselves that the YTM (in totality) really does capture the coupons being reinvested, and we did that by taking our coupons each year, compounding them at the YTM rate for the relevant number of years left until maturity, tacking on the principal we're getting at maturity, and then summing up the proceeds.

Then, by discounting the total proceeds at maturity by the yield to maturity raised to the number of years until maturity, we also find the current price. Thereby proving that YTM is inclusive of the reinvested coupons.

Note: Remember we're saying here that we're taking the coupons immediately upon getting them and reinvesting them at the YTM rate for however many years there are until maturity – so all our cash flows are effectively arriving only at the maturity date.

Anyway, by undertaking this strategy we'll get the following…

Total Proceeds Calculation

	Year 1	Year 2	Year 3	Year 4	Year 5
Coupons	10.00	10.00	10.00	10.00	10.00
Coupon Reinvestment Earnings	8.19	5.66	3.49	1.61	-
Principal	-	-	-	-	100.00
Total Proceeds	**18.19**	**15.66**	**13.49**	**11.61**	**110.00**

Total Proceeds at Maturity	**168.94**
PV of Proceeds at Maturity	**80.00**

So, in total we've found that at maturity we'll have $168.94 of proceeds. If we strip out the price we paid for the bond ($80) that means we have $88.94 of what we'll call "excess proceeds". In other words, the proceeds at maturity above and beyond just recouping what we paid initially for the bond.

The breakdown of these excess proceeds can be found by just looking at the table above: we'll have $50 in coupons, $18.94 of coupon reinvestment earnings, and $20 in price appreciation (since we bought at $80 but are assuming we're getting back par).

Therefore, we can say that the coupons earned represent 50 / 88.94 = 56.22% of these excess proceeds, while the reinvestment earnings represent 18.94 / 88.94 = 21.30% and the price appreciation amount represents 20 / 88.94 = 22.49%.

Now if we multiply the percent of each of these components by the YTM we'll find that 9.07% of the YTM is attributable to the coupons, while 3.43% is attributable to the reinvestment earnings and 3.63% is attributable to the price appreciation.

Putting this all together, we know our current yield (C / P) will just be 10 / 80 or 12.5% and what we've found above is that the portion of the YTM being driven by our coupons (9.07%) and reinvestment earnings (3.43%) is also equal to 12.5%. Therefore, the current yield is reflecting both the coupons earned and their reinvestment at the YTM rate.

Excess Proceeds (Above Purchase Price)	
Total Coupons	50.00
Total Reinvestment Earnings	18.94
Total Price Appreciation	20.00
Total Excess Proceeds	**88.94**

Excess Proceeds Contributions	
Total Coupons	56.22%
Total Reinvestment Earnings	21.30%
Total Price Appreciation	22.49%

YTM Contributions	
Total Coupons	9.07%
Total Reinvestment Earnings	3.43%
Total Price Appreciation	3.63%
Total YTM	**16.13%**

Current Yield Check	
Current Yield	12.5%
YTM Coupons + Reinvestment	12.5%

Note: If you want to take this one more step, we know that a bond with five years to maturity, a 10% coupon, and a 16.125% YTM will be worth $80. Based on our YTM contribution findings above, we'd expect the bond to appreciate by $2.904 next year (3.63% of $80). Punching in 82.904 (80 + 2.904) to a YTM calculator will show you that's the case, or you can check with our quick-and-dirty method we discussed before…

Assumptions	
Maturity	5.0
Coupon	10.0%
Yield	16.1%
Price	80.0

	Year 1	Year 2	Year 3	Year 4	Year 5
Price	80.00	82.90	86.27	90.18	94.72
Plus: Yield	12.90	13.37	13.91	14.54	15.28
Less: Coupon	(10.00)	(10.00)	(10.00)	(10.00)	(10.00)
Final Price	**82.90**	**86.27**	**90.18**	**94.72**	**100.00**
Appreciation Spread (AS)	*2.90*	*3.37*	*3.91*	*4.54*	*5.28*
Current Yield (C / P)	12.5%	12.1%	11.6%	11.1%	10.6%
App. Spread Yield (AS / P)	3.6%	4.1%	4.5%	5.0%	5.6%
Yield to Maturity	**16.1%**	**16.1%**	**16.1%**	**16.1%**	**16.1%**

Let's say that we have a bond maturing in four years with a 5% coupon and a YTM of 10%. Can you tell me, without doing any real math, what the price roughly is?

Alright, this is a more interview-style question that can crop up every now and then. But hopefully after going through the prior question where we found the YTM in a more "intuitive" way you have an inkling of what to do here (there are a few different hacks you could do to get an answer, but we'll focus on answering based more on "first principles").

Because in that prior question we realized that we could get a rough estimate of the YTM by adding up two components: the current yield and the yield informed by our annual principal pickup. This didn't get us an exact figure, as using our little shortcut to find the yield contribution from the annual principal pickup systematically *overstated* how much price appreciation was occurring, but it got us into the right ballpark.

In this question we can take a somewhat similar approach (philosophically) but we're obviously starting from the other end of the pipeline: instead of needing to figure out the YTM based on the current price, we need to find the current price based on the YTM.

So, to get back to basics, we know that if a bond has a coupon of 5%, along with a YTM of 5%, then the bond will be trading at par. But, since the YTM here is substantially higher than the coupon rate, we know that the price of the bond must be significantly lower than par. Likewise, we know from our prior analysis that this higher YTM is inherently reflecting the amount the bond needs to return each year in order to get it back to par at maturity.

Therefore, we'll begin by making the simplifying assumption that the excess yield here (e.g., the spread between the YTM and the coupon rate, in this case 5%) is entirely attributable to the appreciation of the bond. So, we'll say that the bond needs to appreciate in value by $5 each year and, since there are four years to maturity, our initial estimate for the price of the bond will be $100 - $20 or $80.

Now we should know, intuitively, that this simplifying assumption is systemically *understating* the actual price of the bond for a few major reasons.

First, we know that if a bond is trading below par, the right way to think about the yield contribution stemming from our coupon component is by using the current yield (C / P). However, since we're saying the yield contribution of our principal pickup is 5%, we're conversely saying that the yield contribution from our coupon component is 5%.

But we know that can't be right, as we're effectively treating the yield contribution from our coupon component as if the bond is trading at par (e.g., a current yield equal to 5 / 100 = 5%) when we *know* that the bond is per se trading somewhere well below par. So, put another way, we're systematically understating the yield contribution from our coupons – which should be intuitively obvious as we know we're reinvesting our coupons at the YTM rate, and we've been told in the question that's higher than the coupon rate!

Second, we know from our prior analysis that if a bond is trading below par the amount it will increases by year-to-year isn't constant. Instead, the amount of price appreciation the bond experiences will increase each year. (As an aside, this is why we know that our "intuitive" approach to the question where we had to find the YTM involved us per se *overstating* the amount of yield coming from our principal pickup.)

Anyway, in an interview you can just give the answer of $80, explain how you found it, and then caveat that the real answer would be a bit above $80 due to your simplifying assumption overemphasizing how much of the YTM is being driven by principal pickup.

However, we can still get a better approximation of the actual price of the bond without getting into too much real math. Because right now we *know* that $80 is below the actual price of the bond, as we know that we've attributed too little of the 10% in YTM to the coupon component and, conversely, too much to the principal pickup component.

Therefore, let's treat the $20 in principal pickup and the $80 price as just an initial estimate to then hone in on a more accurate (higher) price.

To do this, we'll try to backout a better breakdown of the YTM components. So, we'll start by finding our current yield (C / P) using our little estimated price which will just be 5 / 80 or 6.25%. Since we've been told in the question that the *actual* YTM is 10%, we can then just back out the implied appreciation spread (10% - 6.25% = 3.75%).

Given this, we'll say there will be, on average, $3.75 in principal pickup per year and, since there are four years to maturity, there will be $15.00 in total principal pickup. Therefore, the implied price of the bond is 85.

This is not only quite a different answer from where we started ($80) but is also quite close to the actual answer of $84.15. How close of an approximation we ended up getting is purely attributable from starting from a reasonable price estimate ($80) and, as a consequence, getting a much more reasonable estimate for the amount of the YTM being driven by our coupons.

To the right is a little breakdown of how we got here…

Needless to say, in an interview you just need to give a guesstimate, as we did above, and then briefly explain your rationale.

Because your interviewer isn't looking for precision, they're looking for you to show a reasonably intuitive understanding and provide an answer that's roughly in the right ballpark.

Assumptions	
Maturity	4.00
Coupon	5.00%
Yield	10.00%
Yield - Coupon Spread	5.00%
Par Value	100.00
(-) Estimated Principal Pickup	20.00
Estimated Starting Price	**80.00**
Implied Current Yield	6.25%
Implied App. Spread Yield	3.75%
Implied Principal Pickup	**15.00**
Implied Starting Price	**85.00**

Note: The strategy used above will begin to become less reliable at extremes. For example, if the YTM is 20% then, using our little strategy for arriving at an estimated starting price, we'd be assuming there's $15 in price appreciation per year, so the estimated starting price would be $40. Then, after making our adjustments to get to the implied starting price, we would arrive at $70. But the real starting price is actually $61.16. So, our implied starting price is still much closer to the real price but it's not nearly as accurate as when the coupon and yield aren't too far apart.

The reason our little strategy begins to breakdown is straight-forward: when there's a huge spread between the yield and the coupon rate, as in this hypothetical, the real starting price will be so low that the amount of price appreciation needed is significantly less than what our simplifying assumption would dictate (e.g., if a bond matures in four years with a 5% coupon and a 20% YTM the initial price would be $61.16, and it would only increase to $68.40 next year when there are three years to maturity – an increase of just $7.24 whereas our simplifying assumption had us assuming it would increase $15!).

With all that said, in an actual interview you'll never be given such extreme spreads between the yield and the coupon rate as it'd defeat the purpose of the question. Instead, the yield and coupon rate will be close enough together that you can use our little strategy above where both the estimated starting price and the implied starting price will get you near the real price – just with the latter getting you quite a bit closer than the former.

Note: As illustrated in the preamble, we could find the real price here in a few seconds, assuming we had access to Excel, by just taking the cash flows we're getting (e.g., $5 in coupons a year, $100 at maturity) and discounting them by the YTM rate we were given. But the point of the question isn't to get you to give a perfectly precise answer – rather, it's to see if you can give an answer somewhere in the right ballpark using your "intuition" and, more importantly, explain how you arrived at the answer.

Assumptions	
Maturity	4.0
Coupon	5.0%
Yield	10.0%
Price	84.15

PV Cash Flow Method	**Year 1**	**Year 2**	**Year 3**	**Year 4**
Coupons	5.00	5.00	5.00	5.00
Principal	-	-	-	100.00
Total Cash Flows	**5.00**	**5.00**	**5.00**	**105.00**
PV of Total Cash Flows	**4.55**	**4.13**	**3.76**	**71.72**
PV if C = YTM	**4.76**	**4.54**	**4.32**	**86.38**

PV of Annual Cash Flows	**84.15**
PV of Cash Flows if C = YTM	**100.00**